

Homework #1

Entropy, Relative Entropy and Mutual Information

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw1.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw1.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW1.1 (C&T 2.1) Coin flips. A fair coin is flipped until the first head occurs. Let X denote the number of flips required.

(a) Find the entropy $H(X)$ in bits. The following expressions may be useful:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \quad \sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}.$$

(b) A random variable X is drawn according to this distribution. Find an “efficient” sequence of yes–no questions of the form, “Is X contained in the set S ?” Compare $H(X)$ to the expected number of questions required to determine X .

Problem HW1.2 (C&T 2.2) Entropy of functions. Let X be a random variable taking on a finite number of values. What is the (general) inequality relationship of $H(X)$ and $H(Y)$ if

(a) $Y = 2^X$?

(b) $Y = \cos X$?

Problem HW1.3 (C&T 2.5) Zero conditional entropy. Show that if $H(Y|X) = 0$, then Y is a function of X [i.e., for all x with $p(x) > 0$, there is only one possible value of y with $p(x, y) > 0$].

Problem HW1.4 (C&T 2.8) Drawing with and without replacement. An urn contains r red, w white, and b black balls. Which has higher entropy, drawing $k \geq 2$ balls from the urn with replacement or without replacement? Set it up and show why. (There is both a difficult way and a relatively simple way to do this.)

Problem HW1.5 (C&T 2.12) Example of joint entropy. Let $p(x, y)$ be given by

$X \setminus Y$	0	1
0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	$\frac{1}{3}$

Find:

(a) $H(X)$, $H(Y)$.

- (b) $H(X | Y), H(Y | X)$.
- (c) $H(X, Y)$.
- (d) $H(Y) - H(Y | X)$.
- (e) $I(X; Y)$.
- (f) Draw a Venn diagram for the quantities in parts (a) through (e).

Problem HW1.6 (C&T 2.14) Entropy of a sum. Let X and Y be random variables that take on values $x_1, x_2, \dots, x_r$ and $y_1, y_2, \dots, y_s$, respectively. Let $Z = X + Y$.

- (a) Show that $H(Z|X) = H(Y|X)$. Argue that if X, Y are independent, then $H(Y) \leq H(Z)$ and $H(X) \leq H(Z)$. Thus, the addition of *independent* random variables adds uncertainty.
- (b) Give an example of (necessarily dependent) random variables in which $H(X) > H(Z)$ and $H(Y) > H(Z)$.
- (c) Under what conditions does $H(Z) = H(X) + H(Y)$?

Problem HW1.7 (C&T 2.18) World Series. The World Series is a seven-game series that terminates as soon as either team wins four games. Let X be the random variable that represents the outcome of a World Series between teams A and B; possible values of X are AAAA, BABABAB, and BBBA AAAA. Let Y be the number of games played, which ranges from 4 to 7. Assuming that A and B are equally matched and that the games are independent, calculate $H(X)$, $H(Y)$, $H(Y|X)$, and $H(X|Y)$.

Problem HW1.8 (Challenge; Optional) Mutual information under a linear transformation. Consider arbitrary random variables X_1, X_2 , and

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} + \begin{pmatrix} N_1 \\ N_2 \end{pmatrix}$$

where the matrix elements $[a_{ij}]$ are arbitrary non zero constants and N_1 and N_2 are independent random variables. Let's denote $\mathbf{X} := \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$. Prove or disprove $I(\mathbf{X}; Y_1, Y_2) \leq I(\mathbf{X}; Y_1) + I(\mathbf{X}; Y_2)$.

Problem HW1.9 Two notations for mutual information. $I(X_1, X_2; Y)$ and $I(\mathbf{X}; Y)$. Are they different?

Problem HW1.10 Joint distribution of U and V (HW#0 revisited). Recall the HW#0 problem on the joint distribution of U and V .

- (a) For the first case where $p_1 = 0.1$ and $p_2 = 0.2$, find the following measures: $H(U)$, $H(V)$, $H(U|\theta_1)$, $H(V|\theta_2)$, $H(U|V)$, $H(V|U)$, $H(U, V)$, $I(U; V)$, $I(U; \theta)$, $I(V; \theta)$.
- (b) Repeat for $p_1 = 0.01$ and $p_2 = 0.02$.
- (c) Note there is a notable change in $I(U; V)$ between (a) and (b). Describe this change and make qualitative statements explaining the change. What would happen to $I(U; V)$ when p_1 and p_2 approach zero? What would happen if they both approach $1/2$?

Note: θ_1 and θ_2 here are supposed to be equal to e_1 and e_2 of HW#0, respectively.

Problem HW1.11 *Convexity without calculus.* Showing the convexity of $f(x) = e^x$ by calculus is easy: take the derivative twice and show it is positive everywhere. Now prove the convexity of f using the general convexity-proving technique learned in this chapter (the definition $p_1f(x_1) + p_2f(x_2) \geq f(p_1x_1 + p_2x_2)$ directly).

Problem HW1.12 *Information measures on a noisy pair.* Recall the HW#0 problem on the joint distribution of $U = \theta + e_1 \pmod{2}$ and $V = \theta + e_2 \pmod{2}$.

- (a) For $P_\theta = 0.6$, $p_1 = 0.1$, $p_2 = 0.2$, find $H(U)$, $H(V)$, $H(U|e_1)$, $H(V|e_2)$, $H(U|V)$, $H(V|U)$, $H(U, V)$, $I(U; V)$, $I(U; \theta)$, $I(V; \theta)$.
- (b) Repeat for $p_1 = 0.01$ and $p_2 = 0.02$.
- (c) Describe the change in $I(U; V)$ between (a) and (b) and explain it qualitatively. What happens as $p_1, p_2 \rightarrow 0$? As $p_1, p_2 \rightarrow \frac{1}{2}$?